

Kelsey Frewin

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EDUCATION AND ACADEMIC POSITIONS

Senior Research Associate (2023 – 2026) University of East Anglia

With Prof Teodora Gliga (PI) and Prof Marie Smith (Co-I, Birkbeck)

- Applying EEG multivariate pattern analysis to decode infants' conceptual knowledge (Frewin et al., in prep)
- Invited to lead a book chapter on multivariate EEG methods for Springer Nature
- Leading work with deaf infants born to hearing parents on language access, category learning, and cognitive development, including an opinionated review paper (Frewin et al., under revision)
- Contributed to development of, and data collection for, a UK-based survey on the lived experiences of parents with deaf children (Mercure et al., under revision)
- Co-organising the Early Development & Deafness Network — an international, BSL/ASL-accessible forum uniting researchers, clinicians, teachers, and the deaf community
- Lecturing at undergraduate and postgraduate level on cognitive development in deaf infants and eye-tracking methods in developmental science (2024–2026)

PhD Developmental Cognitive Neuroscience (No Corrections; 2019-2023) Cardiff University

With Drs Sarah Gerson, Ross Vanderwert, and Chiara Gambi

- Established the earliest evidence of verb understanding in during infancy. Used EEG and event-related potentials to demonstrate that 10-month-olds recognise when verbs mislabel familiar actions (Cortex, 2026)
- Demonstrated tight links between infants' vocabulary size and motor development, above and beyond effects of age (Infancy, 2025)

Research Assistant (2021-2022) Cardiff University

With Dr Chiara Gambi

- Eye-tracking and language assessments with 4-year-olds

Research Assistant (2017-2019) Cardiff University

With Drs Sarah Gerson, Ross Vanderwert, and Travis Proulx

- Collected EEG/ERP and eye-tracking data from infants and behavioural data with 3–8-year-olds
- Managed the REDCap database and survey system at Cardiff University, providing university-wide training and support to research teams
- Contributed to parent-survey research assessing parents' usage and beliefs around pacifier usage (Mitev et al., 2024, *Pediatric Research*)

MSc Cognitive Neuroscience (Distinction; 2015-2016) University of Sussex

With Prof Jessica Horst

BSc Psychology (First Class; 2011-2014) Aberystwyth University

TEACHING EXPERIENCE

Lecturing University of East Anglia (2024-2026)

- 3rd Year UG Psychology BSc - Cognitive development in deaf infants
- Developmental Science MSc – Eye-tracking methods in developmental science

Lead Postgraduate Teaching Assistant Cardiff University (2019-2021)

TRAINING COURSES

- **3rd International Workshop on Naturalistic Experimentation of Child Development 2026** (Naturalistic methods in developmental science) Birkbeck College, University of London, UK
- **33rd EEGLAB Summer School Workshop 2023** (EEG processing and analysis software) Aspet, France
- **Neurocognitive Methods for Infant and Toddler Research Summer School 2017** Utrecht University, Netherlands

PUBLICATIONS (* indicates first authorship authorship)

Invited Chapter:

Frewin, K. L.*, Watson, E. A. G., Smith, F., Ewing, L., Santiago, A., Mares, I., & Smith, M. L. (in press). *A Guide to EEG Decoding: From experiment design to Multivariate Pattern Analysis*. In F. F. Santos. (Ed.), *Event-Related Potentials: Methods for reproducible research*. Springer Nature.

Published:

Frewin, K. L.*, Vanderwert, R. E., Gambi, C., Renoult, L., & Gerson, S. A. (2026). Electrophysiological evidence of infants' understanding of verbs. *Cortex*, 196, 41-60. <https://doi.org/10.1016/j.cortex.2025.12.009>

Frewin, K. L.*, Gerson, S. A., Vanderwert, R. E., & Gambi, C. (2024). Parent-Reported Relations Between Vocabulary and Motor Development in Infancy: Differences Between Verbs and Nouns. *Infancy*, 30(1), e12638. <https://doi.org/10.1111/inf.12638>

Mitev, K., **Frewin, K. L.**, Augustinova, M., Niedenthal, P. M., Rychlowska, M., & Vanderwert, R. E. (2024). The who, when, and why of pacifier use. *Pediatric Research*, 97, 2282-2287. <https://doi.org/10.1038/s41390-024-03540-6>

Frewin, K. L.*, McEwen, E., Gerson, S. A., Bekkering, H., & Hunnius, S. (2019). What is going on in babies' brains when they learn to do something? *Frontiers for Young Minds*, 7, 44. <https://doi.org/10.3389/frym.2019.00044>

Submitted:

Mercure, E., Butt, T., **Frewin, K. L.**, Mejia, E., & Althaus, N. (under minor revision). Lived Experiences of Deaf Children's Parents: a UK-based Survey.

Frewin, K. L.*, Althaus, A., Samuelson, L., & Gliga, T. (under minor revision). What comes first, the [concept] CHICKEN or the [word] chicken? The developmental interplay between conceptual and language learning.

In Prep:

Frewin, K. L.*, Smith, M. L., Archer J., Mejia, E., & Gliga, T. (in prep). Decoding infants' category knowledge using EEG Multivariate Pattern Analysis.

Archer, J., **Frewin, K. L.**, Samuelson, L., & Gliga, T. (in prep). When Do Young Children Know that Two Objects are the Same? A Scoping Review of Visual Matching Paradigms.

Bray, E.*, **Frewin, K. L.***, Gliga, T., Hall, M., Mejia, E., & Mercure, E. (in prep). Advances in Assessing Multimodal Multilingual Input and Communicative Development for Deaf and Sign-Exposed Infants: A Multiple Case Study Approach

Paul, K., Wacker, ..., **Frewin, K. L.** (in prep). EEGMANYANALYSTS: Sources of variance in data analysis routines in EEG research.

CONFERENCES

- Frewin, K. L.**, Smith, M. L., & Gliga, T. (2026). Neural markers of prior category knowledge: Decoding object representations using multivariate approaches to infant EEG. *Symposium presentation at International Congress of Infant Studies, Panama City, Panama.*
- Frewin, K. L.**, Vanderwert, R. E., Gambi, C., Renoult, L., & Gerson, S. A. (2026). Electrophysiological evidence of infants' understanding of verbs. *Symposium presentation at International Congress of Infant Studies, Panama City, Panama.*
- Frewin, K. L.**, Smith, M. L., & Gliga, T. (2025). Decoding infants' existing category knowledge using advanced EEG Multivariate Pattern Analysis. *Oral presentation at Lancaster Conference on Infant and Early Child Development, UK.*
- Frewin, K. L.**, Vanderwert, R. E., Gambi, C., & Gerson, S. A. (2026). Electrophysiological evidence of infants' understanding of verbs. *Poster presented at Lancaster Conference on Infant and Early Child Development, UK.*
- Archer, J., **Frewin, K. L.**, Samuelson, L., & Gliga, T., (2026) Designing an eye-tracking task to measure infants' existing category knowledge. *Poster presented at Lancaster Conference on Infant and Early Child Development, UK.*
- Frewin, K. L.**, Vanderwert, R. E., Gambi, C., & Gerson, S. A. (2026). Electrophysiological evidence of infants' understanding of verbs. *Poster presented at International Congress of Infant Studies, Glasgow, UK.*
- Frewin, K. L.**, Mejia, E., Althaus, N., Pomiechowska, B., & Gliga, T. (2024). Do object labels facilitate category matching in 12-month-old infants? *Symposium presentation at International Congress of Infant Studies, Glasgow, UK.*
- Frewin, K. L.** & Gliga, T. (2024). Get rid of that noise! Automated pre-processing pipelines for infant EEG. *Workshop presentation at Neurodevelopmental Annual Seminar (NDAS), University of East Anglia, Norwich, UK.*
- Frewin, K. L.**, Vanderwert, R.E., Gambi, C., & Gerson, S.A. (2022). Investigating the onset of verb comprehension in young infants. *Oral presentation at International Congress of Infant Studies, Ottawa, Canada.*
- Frewin, K. L.**, Vanderwert, R.E., Gambi, C., & Gerson, S. A. (2021). Assessing early verb comprehension in 10- and 14-month-olds: A proposed eye-tracking and event-related potential study. *Poster presented at Lancaster Conference on Infant and Early Child Development, UK.*
- Frewin, K. L.**, Gambi, C., Vanderwert, R.E., & Gerson, S. A. (2021). Infants' fine motor skill, but not gross, predicts concurrent verb comprehension. *Poster presented at Society for Research in Child Development Biennial Meeting, USA.*
- Frewin, K. L.** (2020). The role of motor experiences in infant verb acquisition. *Cardiff University School of Psychology Postgraduate Conference, Cardiff UK. 2nd Place Prize.*
- Frewin, K. L.**, Gambi, C., Vanderwert, R.E., & Gerson, S. A. (2020). Relation between motor development and verb comprehension in infancy. *International Congress of Infant Studies, Glasgow, UK.*
- Vanderwert, R.E., **Frewin, K. L.**, & Rychlowska, M. (2018). Emotional dummy? Implications of pacifier use for emotion face processing. *International Congress of Infant Studies, Philadelphia, USA.*

FUNDING, SCHOLARSHIPS AND AWARDS

- 2020: Junior Researcher of the Year Award. Cardiff University (£200)
- 2019: Student-Led ESRC Wales DTP PhD Studentship. Cardiff University. (Fully-funded)
- 2018: ESRC Festival of Social Science Event, Lead Applicant. Cardiff University (£900)
- 2015: Chancellor's Masters Scholarship. University of Sussex (£3000)
- 2011: Excellence Scholarship. Aberystwyth University (£2000)

PROFESSIONAL MEMBERSHIP

International Congress of Infant Studies; Cognitive Science Society; Society for Research in Child Development

COMMITTEES AND PANELS

- 2023-present: **Early Career Researcher Committee Member**, International Congress of Infant Studies: <https://infantstudies.org/>
- 2022-present: **Early Career Researcher Group Co-organiser**, University of East Anglia
- 2022-present: **Research Committee, Early Career Researcher Representative**, University of East Anglia
- 2018-2019: **Athena Swan Gender Equality Steering Group**, Cardiff University: <https://www.advance-he.ac.uk/equality-charters/athena-swan-charter>

PEER REVIEW

Journals:

Cognition; Developmental Cognitive Neuroscience; Developmental Psychology; Infancy; Infant Behavior and Development; Journal of Cultural Cognitive Science; Journal of Experimental Child Psychology; Language, Cognition, and Neuroscience; Motivation and Emotion; Neuropsychologia; Scientific Reports

Conferences:

Budapest CEU Conference on Cognitive Development 2025; CogSci 2024; International Congress for Infant Studies 2024; Neurodevelopmental Annual Seminar (NDAS) 2024

PROGRAMMING AND SOFTWARE

Programming languages: R, MATLAB

Statistical software: R, SPSS, JASP

EEG Toolboxes: EEGLAB, ERPLAB, APICE

Presentation Software: PsychoPy, E-Prime

Stimulus Editing: Adobe Photoshop, Pixelmator, GIMP, Audacity, PRAAT, iMovie, FFMPEG

Database Management/Surveys: REDCap, Qualtrics, RippleScience

Video Behavioural Coding: ELAN, DataVyu

EVENT ORGANISING AND PUBLIC ENGAGEMENT

- 2023-Ongoing: **Early Development & Deafness Network (online)**, Co-Organiser: Twice yearly international meetings bringing together professionals interested in early cognitive and communicative development of deaf infants and toddlers. It brings together members of the deaf community, teachers, clinicians and researchers; accessible in English, British Sign Language and American Sign Language.
- 2024: **Neurodevelopmental Annual Seminar** (University of East Anglia), Co-organiser: An annual conference that aims to bring together academics and established researchers, as well as post-graduates, post-doctoral researchers and early career researchers in neurodevelopmental conditions and neurodiversity. Focused on organising the EEG for developmental populations workshop.
- 2018: **Stepping into the Minds of Children: An ESRC Festival of Social Science Event**, Lead Organiser: A funded public-engagement event for parents to learn about recent developmental science and engage with developmental scientists from Cardiff University Centre for Human Developmental Science.

WIDENING ACCESS, EQUALITY AND DIVERSITY

- 2021: **Social Mobility Network, Career Mentor and Foundation Futures Day Speaker:** The Social Mobility Foundation is a charity which supports social mobility for young people. SMF provides opportunities, and networks of support for 16-17 year olds who are unable to get them from their schools or families. <https://www.socialmobility.org.uk/>
- 2019: **Nuffield Research Placement Provider and Supervisor:** The Nuffield Foundation provides research placements for students from low-income households and without family history of going to university: <https://www.nuffieldfoundation.org/impact/improving-access-to-university-through-nuffield-research-placements>

REFERENCES

Available upon request